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Semiconductor Package and Method

Abstract

A semiconductor structure includes a first redistribution structure, a first local interconnect component disposed on the first redistribution structure, and a first interconnect structure over a second side of the first local interconnect component. The first local interconnect component includes a first plurality of redistribution layers. The first plurality of redistribution layers includes a first plurality of conductive features on a first side of the first local interconnect component. Each of the first plurality of conductive features are coupled to respective conductive features of the first redistribution structure. The first interconnect structure includes a second plurality of conductive features and a third plurality of conductive features. The second plurality of conductive features are electrically coupled to the third plurality of conductive features through the first local interconnect component.

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Background/Summary

CLAIM TO PRIORITY [0001] This application is a continuation of and claims priority to U.S. patent application Ser. No. 18/745,613, filed on Jun. 17, 2024, and entitled “Semiconductor Package and Method,” which is a divisional of and claims priority to U.S. patent application Ser. No. 17/412,966, filed on Aug. 26, 2021, now U.S. Pat. No. 12,051,650 issued Jul. 30, 2024, and entitled “Semiconductor Package and Method,” each application is incorporated herein by reference.

BACKGROUND

[0002] The semiconductor industry has experienced rapid growth due to ongoing improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, improvement in integration density has resulted from iterative reduction of minimum feature size, which allows more components to be integrated into a given area. As the demand for shrinking electronic devices has grown, a need for smaller and more creative packaging techniques of semiconductor dies has emerged. An example of such packaging systems is Package-on-Package (POP) technology. In a PoP device, a top semiconductor package is stacked on top of a bottom semiconductor package to provide a high level of integration and component density. PoP technology generally enables production of semiconductor devices with enhanced functionalities and small footprints on a printed circuit board (PCB).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 illustrates a cross-sectional view of a package component in accordance with some embodiments.

[0005] FIG. 2 illustrates a detailed view of a portion of the cross-sectional view of FIG. 1 in accordance with some embodiments.

[0006] FIG. 3 illustrates a cross-sectional view of another package component in accordance with some embodiments.

[0007] FIG. 4 illustrates a detailed view of a portion of the cross-sectional view of FIG. 3 in accordance with some embodiments.

[0008] FIGS. 5A and 5B illustrates plan views of package components in accordance with some embodiments.

[0009] FIGS. 6 through 12 illustrate cross-sectional views of intermediate steps during a process for forming a local interconnect component in accordance with some embodiments.

[0010] FIGS. 13 through 20 and 23 through 24 illustrate cross-sectional views of intermediate steps during a process for forming a package component in accordance with some embodiments.

[0011] FIG. 21 illustrates a planar view of the layout of package regions on a wafer substrate in accordance with some embodiments.

[0012] FIG. 22 illustrates a planar view of the layout of package regions on a panel substrate in accordance with some embodiments.

DETAILED DESCRIPTION

[0013] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0014] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0015] Embodiments discussed herein may be discussed in a specific context, namely a package component which has one or more integrated circuit dies. In some embodiments, the package component is a system-on-integrated-substrate (SoIS) package. The package component includes a double-sided local interconnect component embedded in a redistribution structure. The embedded double-sided local interconnect component may provide electrical connection between the integrated circuit dies. The embedded local interconnect component may increase the communication bandwidth between the integrated circuit dies while maintaining low contact resistance and high reliability. The double-sided local interconnect connections may achieve high signal integrity and power integrity, and may enable power distribution networks to be managed through the two-sided connection without a significant voltage drop.

[0016] The redistribution structure is connected to the integrated circuit dies and provides electrical connection between the integrated circuit dies and a core substrate and/or between the integrated circuit dies. The core substrate is additionally connected to a set of external conductive features. In such a manner, the integrated circuit dies are electrically connected to the core substrate, and ultimately to the external conductive features, through the core substrate and the redistribution structure.

[0017] In accordance with some embodiments, the redistribution structure, the embedded double-sided local interconnect component, the core substrate, and the integrated circuit dies may be individually fabricated and tested prior to assembling the completed package component. This further increases component and board level reliability.

[0018] Due to the increased communication bandwidth between the integrated circuit dies provided by the double-sided local interconnect components, an interposer is not required between the integrated circuit dies and the redistribution structure. By removing the need for an interposer, the

warpage mismatch between the integrated circuit package (including the integrated circuit dies) and the core substrate package (including the core substrate and the redistribution structure) is reduced because the coefficient of thermal expansion (CTE) mismatch between these two package structures is reduced.

[0019] The use of double-sided local interconnect components formed without adhesives such as die attach films may allow the reliability window to be enlarged. Omitting a grinding process step to remove the adhesive may reduce the formation of compound cracks between attached integrated circuit dies, improving the reliability of the device.

[0020] FIG. 1 illustrates a cross-sectional view of a singulated package component **100** in accordance with some embodiments. FIG. 2 illustrates a detailed view of region **1000** of the cross-sectional view of FIG. 1, in accordance with some embodiments. The singulated package component **100** includes a semiconductor device (e.g., an integrated circuit package **500**), a redistribution structure **200** having one or more redistribution layers, a core substrate **300**, and external connectors **620**, among other elements. The integrated circuit package **500** may include one or more dies, such as a logic die (e.g., central processing unit (CPU), graphics processing unit (GPU), system-on-a-chip (SoC), application processor (AP), microcontroller, etc.), a memory die (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, etc.), a power management die (e.g., power management integrated circuit (PMIC) die), a radio frequency (RF) die, a sensor die, a micro-electro-mechanical-system (MEMS) die, a signal processing die (e.g., digital signal processing (DSP) die), a front-end die (e.g., analog front-end (AFE) dies), the like, or combinations thereof. In some embodiments, the semiconductor device may be an integrated circuit die.

[0021] The integrated circuit package **500** may include a plurality of integrated circuit dies. As shown, the integrated circuit package **500** includes one or more logic dies **512**, one or more memory dies **514**, and one or more input/output (I/O) dies **516** (not shown in FIG. 1, but see FIGS. 5A and 5B) for illustrative purposes. The integrated circuit dies may be formed in one or more wafers, which may include different device regions that are singulated in subsequent steps. The integrated circuit dies may be packaged with other similar or different integrated circuit dies using known manufacturing techniques.

[0022] In some embodiments, one or more of the integrated circuit dies **512**, **514**, and **516** may be stacked devices that include multiple semiconductor substrates. For example, the memory die **514** may be a memory device such as a hybrid memory cube (HMC) module, a high bandwidth memory (HBM) module, or the like that includes multiple memory dies. In such embodiments, the memory die **514** includes multiple semiconductor substrates interconnected by through-substrate vias (TSVs). Each of the semiconductor substrates may (or may not) have an interconnect structure. An encapsulant **520** may surround the integrated circuit dies **512**, **514**, and **516**.

[0023] The dies **512**, **514**, and **516** have bond pads **518** that are bonded to the conductive connectors **188**. In some embodiments, the bond pads **518** are made of a conductive material and may be similar to the conductive lines (see, e.g., conductive lines **110**) described below.

[0024] Conductive connectors **188** provide electrical connection between the redistribution structure **200** and the integrated circuit package **500**. An underfill **610** may be included to securely bond the integrated circuit package **500** to the redistribution structure **200** and provide structural support and environmental protection.

[0025] As discussed in greater detail below, the redistribution structure **200** provides electrical pathing and connection between the integrated circuit package **500** and a core substrate **300** by way of conductive connectors **390**. In some embodiments, the redistribution structure **200** has one or more redistribution layers comprising metallization patterns, comprising, for example, conductive lines **110** and conductive vias **106** and **112**, and dielectric layers **108** and **114** on top and bottom sides of the conductive lines **110**.

[0026] As discussed in greater detail below, the redistribution structure **200** includes one or more

local interconnect components **120**. The local interconnect components **120** provide electrical routing and connection between the integrated circuit dies **512**, **514**, and **516** of the integrated circuit package **500** and may be referred to as interconnecting dies **120** or local interconnect structures **120**. The local interconnect components **120** increase the communication bandwidth between the integrated circuit dies **512-516** while maintaining low contact resistance and high reliability. The low contact resistance and high reliability is at least in part due to a solder-free connection between the embedded local interconnect component and the redistribution structure. As illustrated in FIGS. **1** and **2**, the local interconnect components **120** are connected to conductive vias **112** of the redistribution structure **200** by conductive connectors **238** that may be metal pillars such as e.g. copper pillars.

[0027] Due to the increased communication bandwidth between the integrated circuit dies provided by the local interconnect components, an interposer is not required between the integrated circuit dies and the redistribution structure. By removing the need for an interposer, the warpage mismatch between the integrated circuit package (including the integrated circuit dies) and the core substrate package (including the core substrate and the redistribution structure) is reduced because the coefficient of thermal expansion (CTE) mismatch between these two package structures is reduced.

[0028] The local interconnect components **120** are further electrically connected on their bottom sides to the core substrate **300** by way of conductive connectors **390**. This double-sided connection of the integrated circuit dies **512**, **514**, and **516** to the core substrate **300** may provide high bandwidth communication with lower resistance, which may enable increasing signal and power integrity.

[0029] As discussed in greater detail below, the local interconnect components **120** may be encapsulated by an encapsulant **420** which may be formed of a molding compound, epoxy, or the like, and may be applied by compression molding, transfer molding, or the like. Through vias **118** may extend through the encapsulant **420** adjacent to the local interconnect components **120** and may be electrically coupled to the conductive lines **110** by conductive vias (not illustrated) extending through the dielectric layer **114**.

[0030] As discussed in greater detail below, the local interconnect components **120**, through vias **118**, and encapsulant **420** may be disposed on one or more redistribution layers comprising metallization patterns, comprising, for example, conductive lines **208** and conductive vias **202**, **204**, and **212**, and dielectric layers **206** and **210** on top and bottom sides of the conductive lines **208**. The through vias **118** may be physically and electrically coupled on bottom sides to the conductive vias **204**. Conductive connectors **122** such as metal pillars, e.g. copper pillars, on the bottom side of the local interconnect components **120** may be bonded to the conductive vias **202** by metal-to-metal bonding such as e.g. copper-to-copper bonding. In some embodiments, the local interconnect components **120** are bonded to the conductive vias **202** and the dielectric layer **206** by hybrid bonding.

[0031] The conductive vias **202** and **204** may be electrically coupled to the conductive lines **208**, which may be electrically coupled to conductive vias **212** extending to a bottom surface of the dielectric layer **210**. Conductive pads **214** on a bottom side of the redistribution structure **200** may be electrically coupled to the conductive vias **212**.

[0032] The redistribution structure **200** may be electrically and mechanically attached to the core substrate **300**. The core substrate **300** may include a central core **310**, with conductive vias **320** extending through the central core **310**, and additional optional redistribution structures **340** along opposing sides of the central core **310**. Generally, the core substrate **300** provides structural support for the component package, as well as providing electrical signal routing between the integrated circuit package and the external connectors **620**, which may be physically and electrically coupled to the conductive pads **214**.

[0033] Underfill **380** may be included between the redistribution structure **200** and the core substrate **300** to securely bond the associated elements and provide structural support and

environmental protection.

[0034] FIG. 3 illustrates a cross-sectional view of another singulated package component **100'** in accordance with some other embodiments. FIG. 4 illustrates a detailed view of region **1000'** of the cross-sectional view of FIG. 3, in accordance with some embodiments. The singulated package component **100'** as illustrated in FIGS. 3 and 4 is similar to the singulated package component **100** illustrated in FIGS. 1 and 2, except with an underfill **138** around the conductive connectors **238** between the local interconnect **120** and the dielectric layer **114**, as discussed in greater detail below (see FIGS. 15C and 15D). The material of the underfill **138** may be a liquid epoxy, deformable gel, silicon rubber, a combination thereof. The underfill **138** and the encapsulant **420** are made of different materials. In some embodiments, the dielectric layer **210** (see above, FIGS. 1 and 2) may be replaced with an encapsulant **430**, which may be formed with the same materials as the encapsulant **420** (see above, FIGS. 1 and 2).

[0035] FIGS. 5A and 5B illustrate plan views of the package component in accordance with some embodiments. The embodiment illustrated in FIG. 5A includes two logic dies **512**, four memory dies **514**, two I/O dies **516**, and seven local interconnect components **120**. In this embodiment, a first logic die **512** is connected to a first memory die **514** through a first local interconnect component **120** (see above, FIGS. 1-4), the first logic die **512** is connected to a second memory die **514** through a second local interconnect component **120**, the first logic die **512** is connected to a third memory die **514** through a third local interconnect component **120**, and the first logic die **512** is connected to a fourth memory die **514** through a fourth local interconnect component **120**. A second logic die **512** is connected to a first I/O die **516** through a fifth local interconnect component **120** and the second logic die **512** is connected to a second I/O die **516** through a sixth local interconnect component **120**. In addition, the first logic die **512** and the second logic die **512** are connected together by a seventh local interconnect component **120**.

[0036] The embodiment illustrated in FIG. 5B includes two logic dies **512**, four memory dies **514**, two I/O dies **516**, and one local interconnect component **120**. In this embodiment, the first logic die **512** and the second logic die **512** are connected to each other by a single local interconnect component **120**, and the other dies are connected to each other through redistribution layers in the redistribution structure **200**. Other embodiments may include more or less logic dies **512**, memory dies **514**, I/O dies **516**, and local interconnect components **120**. In some embodiments, each of the integrated circuit dies are connected to each adjacent integrated circuit die by a local interconnect component.

[0037] FIGS. 6 through 10 illustrates various intermediate stages in fabricating a local interconnect component (see FIG. 10), in accordance with some embodiments. The illustrations of the individual features have been simplified in FIGS. 4 through 16 for ease of illustration.

[0038] Referring first to FIG. 6, a carrier substrate **102** is provided, the carrier substrate **102** may be a glass carrier substrate, a ceramic carrier substrate, a semiconductor substrate, such as a bulk semiconductor, or the like. The carrier substrate **102** may be a wafer, such that multiple redistribution structures can be formed on the carrier substrate **102** simultaneously. The carrier substrate **102** may comprise silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The carrier substrate **102** may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. In some embodiments, the carrier substrate **102** may be made up of a ceramic material, a polymer film, a magnetic material, the like or a combination thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. The carrier substrate **102** has an active surface (e.g., the surface facing upwards in FIG. 6), sometimes called a front side, and an inactive surface (e.g., the surface facing downwards in FIG. 6), sometimes called a back side.

[0039] In FIG. 7, a dielectric layer **222** may be formed on the carrier substrate **102**. The bottom

surface of the dielectric layer **222** may be in contact with the top surface of the carrier substrate

102. In some embodiments, the dielectric layer **222** is formed of a polymer, such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), or the like. In other embodiments, the dielectric layer **108** is formed of a nitride such as silicon nitride; an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like; or the like. The dielectric layer **222** may be formed by any acceptable deposition process, such as spin coating, CVD, laminating, the like, or a combination thereof.

[0040] In FIG. **8**, the dielectric layer **222** is then patterned to form openings **224** exposing portions of the top surface of the carrier substrate **102**. The patterning may be formed by an acceptable process, such as a lithographic process including exposing the dielectric layer **222** to light when the dielectric layer **222** is a photo-sensitive material or by etching using, for example, an anisotropic etch. If the dielectric layer **222** is a photo-sensitive material, the dielectric layer **222** can be developed after the exposure.

[0041] In FIG. **9**, a metallization pattern including conductive elements such as conductive lines **226** extending along the major surface of the dielectric layer **222** and conductive vias **122** extending through the dielectric layer **222** to fill the openings **224**. As an example to form the metallization pattern, a seed layer is formed over the dielectric layer **222** and in the openings **224** extending through the dielectric layer **22**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photoresist is then formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the conductive lines **226** of the metallization pattern. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is then formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. The combination of the conductive material and underlying portions of the seed layer form the conductive connectors **122**, such as metal pillars e.g. copper pillars, and conductive lines **226** of the metallization pattern. The photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching.

[0042] In FIG. **10**, a redistribution structure **220** is fabricated by forming additional redistribution layers comprising dielectric layers with embedded metallization patterns comprising conductive lines and vias on the dielectric layer **222** and the conductive lines **226**. The conductive lines and vias are arranged so that the local redistribution structure **220** may be subsequently be singulated into local interconnect components (see below, FIGS. **11-12**). In the illustrated embodiment, five redistribution layers are formed in the redistribution structure **220**, but any suitable number of redistribution layers may be formed, such as two to 10 redistribution layers. The dielectric layers, conductive lines, and vias may be formed using similar methods and materials as the dielectric layer **222**, the conductive connectors **122**, and the conductive lines **226**. The metallization patterns are formed so that conductive vias **236** extending to a top surface of the dielectric layer **234** (on a top surface of the redistribution structure **220**) are electrically coupled to neighboring conductive vias **236** and to conductive connectors **122** extending to the bottom side of the redistribution structure **220**.

[0043] In FIG. **11**, conductive connectors **238** are formed for external connection to conductive vias **236**. The conductive connectors **238** have bump portions on and extending along the major surface

of the dielectric layer **234**, and may have via portions extending into the dielectric layer **234** to physically and electrically couple the conductive vias **236**. As a result, the conductive connectors **238** are electrically coupled to the conductive connectors **122** extending to the bottom side of the redistribution structure **220**. The conductive connectors **238** may be formed of the same material as the conductive vias **236**. In some embodiments, the conductive connectors **238** are metal pillars, e.g. copper pillars, or micro bumps. Neighboring conductive connectors **238** may be separated by a pitch **P1** in a range of about 20 μm to about 80 μm .

[0044] Still referring to FIG. **11**, solder regions **240** are formed on the conductive connectors **238**. The solder regions **240** allow for physical and electrical connection to dies or another package structure. The solder regions **240** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The solder regions **240** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the solder regions **240** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the solder regions **240** comprise metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process.

[0045] Further in FIG. **11**, a singulation process is performed by sawing along scribe lines **250**. The sawing singulates redistribution structure **220** to form multiple singulated local interconnect components **120** (see below, FIG. **12**).

[0046] Optionally, the solder regions **240** may be used to perform chip probe (CP) testing on the local interconnect components **120**. CP testing may be performed on the local interconnect component **120** to ascertain whether the local interconnect component **120** is a known good die (KGD). Thus, only local interconnect components **120**, which are KGDs, undergo subsequent processing are packaged, and dies, which fail the CP testing, are not packaged. After testing, the solder regions **240** may be removed in subsequent processing steps (see below, FIG. **19**).

[0047] FIG. **12** illustrates a singulated local interconnect component **120**. In the illustrated embodiment, the local interconnect component **120** has five redistribution layers. In other embodiments, local interconnect components **120** may have two to 10 redistribution layers. The first redistribution layer of the illustrated local interconnect component **120**, comprising the dielectric layer **222** and conductive connectors **122**, is on a top surface of a remaining portion of the carrier substrate **102**. Conductive connectors **122** extend through the dielectric layer **222** to the top surface of the remaining portion of the carrier substrate **102** and are electrically coupled to conductive lines **226**. The conductive connectors **122** may subsequently be used to couple to other components such as core substrates **300** (see above, FIGS. **1** and **2**). The conductive lines **226** are coupled through intermediate conductive features such as additional conductive vias and lines to conductive lines **232** in a top redistribution layers of the local interconnect component **120**. The conductive lines **232** may be coupled through conductive vias **236** to conductive connectors **238** and solder regions **240**, which may be subsequently used to couple integrated circuit dies **512**, **514**, and **516** to each other and to other components such as a core substrate **300** (see above, FIGS. **1** and **2**). The double-sided connection of the integrated circuit dies **512**, **514**, and **516** to each other and to the core substrate **300** through the local interconnect component **120** may increase the communication bandwidth between the integrated circuit dies while maintaining low contact resistance and high reliability, which may achieve high signal integrity and power integrity, and

may enable power distribution networks to be managed through the two-sided connection without a significant voltage drop.

[0048] In some embodiments, the local interconnect component **120** has a horizontal width in a range of about 2 mm to about 50 mm, and a horizontal length in a range of about 3 mm to about 80 mm.

[0049] FIGS. **13** through **24** illustrates various intermediate stages in fabricating a redistribution structure **200** (see FIG. **23**), in accordance with some embodiments. A first package region **101A** and a second package region **101B** are illustrated where each package region is eventually singulated from other package regions. The illustrations of the individual features have been simplified in FIGS. **13** through **24** for ease of illustration.

[0050] Referring first to FIG. **13**, a carrier substrate **702** is provided. The carrier substrate **702** may be a glass carrier substrate, a ceramic carrier substrate, or the like. The carrier substrate **702** may be a wafer, such that multiple redistribution structures can be formed on the carrier substrate **702** simultaneously. The carrier substrate **702** may have a release layer (not illustrated) over its top surface. The release layer may be formed of a polymer-based material, which may be removed along with the carrier substrate **702** from the overlying structures that will be formed in subsequent steps. In some embodiments, the release layer is an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a light-to-heat-conversion (LTHC) release coating. In other embodiments, the release layer may be an ultra-violet (UV) glue, which loses its adhesive property when exposed to UV lights. The release layer may be dispensed as a liquid and cured, may be a laminate film laminated onto the carrier substrate **702**, or may be the like. The top surface of the release layer may be leveled and be planar within process variations.

[0051] In FIG. **14**, a metallization pattern may be formed on the carrier substrate **702**. As an example to form the metallization pattern, a seed layer is formed over the carrier substrate **702**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, physical vapor deposition (PVD) or the like. A photoresist (not shown) is then formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the metallization pattern to be formed. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. Then, the photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form the metallization pattern.

[0052] Still referring to FIG. **14**, through vias **118** are formed on portions of the metallization pattern. As an example to form the through vias **118**, a photoresist is formed and patterned on the metallization pattern. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to conductive vias. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is formed in the openings of the photoresist and on the exposed portions of the metallization pattern, forming the through vias **118**. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. The photoresist is removed, such as by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Remaining

portions **704**, also referred to as pads **704**, of the metallization pattern upon which through vias **118** are not formed may subsequently be used for attaching local interconnect components **120** (see below, FIGS. **15A-B**).

[0053] In FIGS. **15A** and **15B**, the local interconnect components **120** are bonded to the pads **704** on the carrier substrate **702**. FIG. **15B** illustrates a detailed view of region **1002** of FIG. **15A**. In some embodiments, the local interconnect components **120** may be placed on the carrier substrate **702** using a pick and place process or another suitable process and the solder regions **240** bonded by flip chip bonding process or other suitable bonding process. In some embodiments, the solder regions **240** are reflowed to attach the local interconnect components **120** to the carrier substrate **702** by way of pads **704**.

[0054] FIGS. **15C** and **15D** illustrate another embodiment in which an underfill **138** is disposed on a surface of the local interconnect component **120** around the conductive connectors **238**. FIG. **15D** illustrates a detailed view of region **1004** of FIG. **15C**. The underfill **138**, also shown in FIGS. **3** and **4**, may be formed surrounding the conductive connectors **238** by a capillary flow process or another suitable deposition method between the local interconnect components **120** and the carrier substrate **702**. The underfill **138** may reduce stress on the conductive connectors **238**. The material of the underfill **138** may be a liquid epoxy, deformable gel, silicon rubber, a combination thereof.

[0055] In FIG. **16**, an encapsulant **420**, also referred to as a dielectric material underfill **420**, is formed on and around the through vias **118** and the local interconnect components **120** in accordance with some embodiments. The encapsulant **420** encapsulates the local interconnect components **120** and the through vias **118**. In some embodiments, the encapsulant **420** may be formed of pre-preg, Ajinomoto Build-up Film (ABF), resin coated copper (RCC), molding compound, polyimide, photo-imageable dielectric (PID), epoxy, epoxy molding compound, dispense molding underfill, or the like, and may be applied by compression molding, transfer molding, or the like. The encapsulant may be applied in liquid or semi-liquid form and then subsequently cured. In some embodiments, the encapsulant **420** is formed over the carrier substrate **702** such that the through vias **118** and local interconnect components **120** are buried or covered, and a planarization process is then performed to remove a top portion of the encapsulant **420** and the substrates **102** of the local interconnect components **120** to expose the through vias **118** and the conductive connectors **122** of the local interconnect components **120**. Topmost surfaces of the encapsulant **420**, through vias **118**, and the conductive connectors **122** of the local interconnect components **120** are level (e.g., planar) within process variations after the planarization process. The planarization process may be, for example, a grinding or a chemical-mechanical polish (CMP). In some embodiments, the encapsulant **420** may comprise other materials, such as silicon oxide, silicon nitride, or the like. After the planarization process (if any), the vertical thickness of the local interconnect components measured from a bottom surface of dielectric layer **234** to a top surface of dielectric layer **222** may be in a range of about 10 μm to about 100 μm .

[0056] The local interconnect components **120** provide electrical connection between the subsequently attached integrated circuit dies (e.g., **512**, **514**, and **516**) and other components such as the core substrate **300**. The embedded local interconnect components **120** may increase the communication bandwidth between the integrated circuit dies and the core substrate **300** while maintaining low contact resistance and high reliability. In some embodiments, other components such as an integrated voltage regulator, an integrated passive device, a static random-access-memory, the like, or a combination thereof can also be embedded in a similar manner as the embedded local interconnect component.

[0057] In FIG. **17**, a redistribution structure **270**, also referred to as a back-side redistribution structure **270** or an interconnect structure **270**, comprising redistribution layers **260** and **262** is formed over the encapsulant **420**, through vias **118**, and local interconnect components **120**. Although the embodiments illustrated in accordance with FIG. **17** show the back-side redistribution structure **270** with two redistribution layers, any suitable number of redistribution layers may be

formed, such as from one to twenty redistribution layers. The redistribution layers **260** and **262** may be used to couple the through vias **118** and local interconnect components **120** to subsequently attached components such as e.g. core substrates **300** (see above, FIGS. **1** and **2**).

[0058] The redistribution layer **260** comprises a dielectric layer **206** and conductive vias **202** and **204** extending through the dielectric layer **206** to couple the conductive connectors **122** of the local interconnect components **120** and the through vias **118**, respectively. The dielectric layer **206** may be formed on the encapsulant **420**, the through vias **118**, and the local interconnect components **120**. In some embodiments, the dielectric layer **108** is formed of a polymer, such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), or the like. In other embodiments, the dielectric layer **206** is formed of a nitride such as silicon nitride; an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like; or the like. The dielectric layer **206** may be formed by any acceptable deposition process, such as spin coating, CVD, laminating, the like, or a combination thereof.

[0059] In some embodiments, the conductive vias **202** and **204** are formed with a damascene process in which the dielectric layer **206** is patterned and etched utilizing photolithography techniques to form trenches corresponding to the desired pattern of conductive vias, such as over the through vias **118** and the conductive connectors **122** of the local interconnect components **120**. An optional diffusion barrier and/or optional adhesion layer may be deposited and the trenches may be filled with a conductive material. Suitable materials for the barrier layer includes titanium, titanium nitride, titanium oxide, tantalum, tantalum nitride, titanium oxide, or other alternatives, and suitable materials for the conductive material include copper, silver, gold, tungsten, aluminum, combinations thereof, or the like. In an embodiment, the conductive vias **202** and **204** may be formed by depositing a seed layer of copper or a copper alloy, and filling the trenches by electroplating. A chemical mechanical planarization (CMP) process or the like may be used to remove excess conductive material from a surface of the dielectric layer **206** and to planarize the surface for subsequent processing.

[0060] Still referring to FIG. **17**, a redistribution layer **262** is formed on the redistribution layer **260**. The redistribution layer **262** comprises conductive features such as a conductive lines **208** and conductive vias **212** embedded in a dielectric layer **210**. The conductive lines **208** and conductive vias **212** may couple the conductive vias **202** and **204** to subsequently attached components such as e.g. core substrates **300** (see above, FIGS. **1** and **2**).

[0061] As an example to form the redistribution layer **262**, a seed layer is formed over the dielectric layer **206** and the conductive vias **202** and **204**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photoresist is then formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the conductive lines **208**. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is then formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. The combination of the conductive material and underlying portions of the seed layer form the conductive lines **208**. The photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching.

[0062] Still referring to FIG. **17**, conductive vias **212** are formed on and extending from the conductive lines **208**. As an example to form the conductive vias **212**, a photoresist is formed and

patterned on the conductive lines **208**. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The patterning forms openings through the photoresist to expose the conductive lines **208**, where the openings in the photoresist correspond to the conductive vias **212**. A conductive material is then formed in the openings of the photoresist and on the exposed portions of the conductive lines **208**. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. The conductive material forms the conductive vias **212**. The photoresist is then removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. In some embodiments, the conductive vias **212** have widths in a range of about 2 μm to about 100 μm , such as about 30 μm .

[0063] Further referring to FIG. **17**, a dielectric layer **210** is formed on and around the conductive lines **208** and the conductive vias **212** in accordance with some embodiments. After formation, the dielectric layer **210** surrounds the conductive vias **212** and conductive lines **208**. The dielectric layer **210** and metallization pattern, including conductive vias **212** and conductive lines **208**, form the redistribution layer **262**. In some embodiments, the dielectric layer **210** is formed of a polymer, such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), or the like. In other embodiments, the dielectric layer **210** is formed of a nitride such as silicon nitride; an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like; or the like. The dielectric layer **210** may be formed by any acceptable deposition process, such as spin coating, CVD, laminating, the like, or a combination thereof. The dielectric material may be applied in liquid or semi-liquid form and then subsequently cured to form the dielectric layer **210**. In still other embodiments, the dielectric layer **210** may be replaced by an encapsulant **430** (see above, FIGS. **3** and **4**) formed of pre-preg, Ajinomoto Build-up Film (ABF), resin coated copper (RCC), molding compound, polyimide, photo-imageable dielectric (PID), epoxy, epoxy molding compound, or the like, and may be applied by compression molding, transfer molding, or the like.

[0064] In some embodiments, the dielectric layer **210** is formed over the dielectric layer **206** such that the conductive lines **208** and conductive vias **212** are buried or covered, and a planarization process is then performed on the dielectric layer **210** to expose the conductive vias **212**. Topmost surfaces of the dielectric layer **146** and conductive vias **144** are level (e.g., planar) within process variations after the planarization process. The planarization process may be, for example, a CMP. In some embodiments, the dielectric layer **210** may comprise other materials, such as silicon oxide, silicon nitride, or the like. In some embodiments, the redistribution layers **260** and **262** have a total thickness in a range of about 2 μm to about 50 μm .

[0065] Still referring to FIG. **17**, conductive pads **214** are formed on the dielectric layer **210** and coupled to the conductive vias **212**. External connectors may subsequently be formed on conductive pads **214** to couple to external devices such as e.g. core substrates **300** (see above, FIGS. **1** and **2**). The conductive pads may be formed using similar methods and materials as described above with respect to the conductive lines **208**.

[0066] In FIG. **18**, a carrier substrate de-bonding is performed to detach (or “de-bond”) the carrier substrate **702** from the pads **704**, encapsulant **420**, and through vias **118**. In accordance with some embodiments, the de-bonding includes projecting a light such as a laser light or an UV light on the release layer (not illustrated) over the carrier substrate **702** so that the release layer decomposes under the heat of the light and the carrier substrate **702** can be removed. The structure is then flipped over and placed on another carrier substrate **802** and release layer (not illustrated).

[0067] In FIG. **19**, a planarization such as a grinding or a chemical mechanical polish is performed to remove the pads **704** and solder regions **240** (see FIG. **18**), exposing top surfaces of the conductive connectors **238** of the local interconnect components **120**. Top portions of the encapsulant **420** and through vias **118** may also be removed. After the planarization, top surfaces of

the conductive connectors **238** may be level with top surfaces of the encapsulant **420** and top surfaces of the through vias **118**.

[0068] In FIG. **20**, an interconnect structure **284**, also referred to as a redistribution structure **284**, comprising redistribution layers **280** and **282** is formed over the encapsulant **420**, through vias **118**, and local interconnect components **120**. Although the embodiments illustrated in accordance with FIG. **20** show the interconnect structure **284** with two redistribution layers, any suitable number of redistribution layers may be formed, such as from one to ten redistribution layers. The redistribution layers **280** and **282** may be used to couple the through vias **118** and local interconnect components **120** to subsequently attached components such as e.g. integrated circuit dies **512**, **514**, and **516**.

[0069] The redistribution layer **280** comprises a dielectric layer **114** and conductive vias **112** extending through the dielectric layer **114** to couple the conductive connectors **238** of the local interconnect components **120**. Additional conductive vias (not illustrated) may extend through the dielectric layer **114** to couple the through vias **118**. The dielectric layer **114**, conductive vias **112**, and additional conductive vias (not illustrated) coupled to the through vias **118** may be formed using similar methods and materials as the dielectric layer **206** and the conductive vias **202** and **204** (see above, FIG. **17**). In some embodiments, the conductive vias **112** and additional conductive vias (not illustrated) have widths in a range of about 2 μm to about 50 μm , such as about 10 μm .

[0070] Redistribution layer **282** is formed over redistribution layer **280** and comprises a dielectric layer **108** and a metallization pattern comprising conductive lines **110** and conductive vias **106**. The conductive lines **110** may be formed on and coupled to the conductive vias **112** and the additional conductive vias (not illustrated) coupled to the through vias **118**. The conductive lines **110**, conductive vias **106**, and dielectric layer **108** may be formed using similar methods and materials as the conductive lines **208**, conductive vias **212**, and dielectric layer **210** (see FIG. **17**). In some embodiments, the redistribution layers **280** and **282** have a total thickness in a range of about 2 μm to about 50 μm .

[0071] Still referring to FIG. **20**, under-bump metallizations (UBMs) **264** (sometimes referred to as pads **264**) are formed for external connection to conductive vias **106**. The UBMs **264** have bump portions on and extending along the major surface of the dielectric layer **108**, and may have via portions extending into the dielectric layer **108** to physically and electrically couple the conductive vias **106**. As a result, the UBMs **264** are electrically coupled to the through vias **118** and the local interconnect components **120**. The UBMs **264** may be formed of the same material as the conductive vias **106**.

[0072] Further referring to FIG. **20**, conductive connectors **188** are formed on the UBMs **264**. The conductive connectors **188** allow for physical and electrical connection to dies such as e.g. integrated circuit dies **512**, **514**, and **516** or another package structure. The conductive connectors **188** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **188** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **188** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the conductive connectors **188** comprise metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process. In some embodiments, neighboring conductive connectors are

separated by a pitch in a range of about 20 μm to about 80 μm .

[0073] As discussed above, the redistribution structure **200** may be larger and include multiple package regions, such as the first package region **101A** and second package region **101B**. For example, FIG. **21** illustrates the redistribution structure **200** as illustrated above in FIG. **20** having a circular wafer shape with multiple package regions. In the embodiment shown, four package regions **101A**, **101B**, **101C**, and **101D** are included on the wafer allowing for four final package components to be fabricated on a single wafer and later singulated. Fewer or more package regions may be utilized on a single wafer in other embodiments. Subsequent steps in the process use the redistribution structure **200** on a carrier substrate **802** as the base upon which to continue the fabrication process described in further detail below. As described in further detail below, the individual package regions are singulated by sawing along lines **401** and around the outer edges of package regions **101A**, **101B**, **101C**, and **101D**.

[0074] FIG. **22** illustrates the redistribution structure **200** being manufactured using a panel form fabrication process with multiple package regions. In the embodiment shown, nine package regions **101A** through **101I** are included on the wafer allowing for nine final package components to be fabricated on a single wafer or panel. Fewer or more package regions may be utilized on a single wafer or panel in other embodiments. Subsequent steps in the process use the redistribution structure **200** on a carrier substrate **802** as the base upon which to continue the fabrication process described in further detail below. As described in further detail below, the individual package regions are singulated by sawing along lines **402** and around the perimeter of package regions **101A** through **101I**.

[0075] In FIG. **23**, a singulation process is performed by sawing along scribe line regions, e.g., between the first package region **101A** and the second package region **101B**. The sawing singulates the first package region **101A** from adjacent package regions to form multiple singulated redistribution structures **200**.

[0076] Further referring to FIG. **23**, an integrated circuit package **500** is attached to the redistribution structure **200**. The attachment may be performed before or after singulating individual package regions of the redistribution structure **200**. The integrated circuit package **500** may include a plurality of integrated circuit dies such as one or more logic dies **512**, one or more memory dies **514**, and one or more input/output (I/O) dies **516** (not shown in FIG. **23**, but see FIGS. **5A** and **5B**) for illustrative purposes. The dies **512**, **514**, and **516** have bond pads **518** that are bonded to the conductive connectors **188**. In some embodiments, the bond pads **518** are made of a conductive material and may be similar to the conductive lines (see, e.g., conductive lines **110**) described above. In some embodiments, the dies **512**, **514**, and **516** may be placed on the redistribution structure **200** using a pick and place process or another suitable process and the conductive connectors **188** bonded by flip chip bonding process or other suitable bonding process. In some embodiments, the conductive connectors **188** are reflowed to attach the dies **512**, **514**, and **516** to the redistribution structure **200** by way of the bond pads **518**. The dies **512**, **514**, and **516** may be coupled to the redistribution structure **200** such that neighboring dies **512**, **514**, and **516** are coupled to each other through local interconnect components **120** (see above, FIGS. **5A** and **5B**). The local interconnect components **120** provide electrical routing and connection between the integrated circuit dies **512**, **514**, and **516** of the integrated circuit package **500** and increase the communication bandwidth between the integrated circuit dies **512-516** while maintaining low contact resistance and high reliability.

[0077] In some embodiments, an underfill **610**, as shown in FIG. **1**, is formed surrounding the conductive connectors **188** between the integrated circuit package **500** and the redistribution structure **200**. The underfill **610** may reduce stress and protect the joints resulting from the reflowing of the conductive connectors **188**. The underfill **610** may be formed by a capillary flow process after the integrated circuit package **500** is attached, or may be formed by a suitable deposition method. In some embodiments, a single layer of underfill **610** is formed beneath

multiple adjacent devices, and further subsequent underfills (not shown) or encapsulants (not shown) may be formed beneath and/or around additional devices placed on top of the singulated package component **100** (see below, FIG. **24**).

[0078] In some embodiments, an encapsulant **520** may be formed around the integrated circuit dies **512**, **514**, and **516**. The encapsulant **520** may be formed of a molding compound, epoxy, or the like, and may be applied by compression molding, transfer molding, or the like. The encapsulant **520** may be applied in liquid or semi-liquid form and then subsequently cured. The encapsulant **520** may be formed over the structure such that integrated circuit dies **512**, **514**, and **516** are buried or covered.

[0079] FIG. **24** illustrates a singulated package component **100** formed by attaching the integrated circuit package **500** and redistribution structure **200** to another package, such as e.g. a core substrate **300**. A carrier substrate de-bonding is performed to detach (or “de-bond”) the carrier substrate **802** from the integrated circuit package **500** and redistribution structure **200**. In accordance with some embodiments, the de-bonding includes projecting a light such as a laser light or an UV light on the release layer (not illustrated) over the carrier substrate **802** so that the release layer decomposes under the heat of the light and the carrier substrate **802** can be removed. The structure is then flipped over and placed on another carrier, such as a tape (not illustrated).

[0080] Still referring to FIG. **24**, a core substrate **300** is illustrated and bonded the redistribution structure **200**. Utilizing the core substrate **300** has the advantage of having the core substrate **300** being manufactured in a separate process. In addition, because core substrate **300** is formed in a separate process, it can be tested separately so that a known good core substrate **300** is used. For example, in some embodiments, the core substrate **300** may be individually or batch tested, validated, and/or verified prior to bonding the core substrate **300** to the redistribution structure **200**.

[0081] The core substrate **300** may be, e.g., an organic substrate, a ceramic substrate, a silicon substrate, or the like. The conductive connectors **390** are used to attach the core substrate **300** to the redistribution structure **200**. Attaching the core substrate **300** may include placing the core substrate **300** on the redistribution structure **200** and reflowing the conductive connectors **390** to physically and electrically couple the core substrate **300** and the redistribution structure **200**.

[0082] Before being attached to the redistribution structure **200**, the core substrate **300** may be processed according to applicable manufacturing processes to form redistribution structures in the core substrate **300**. For example, the core substrate **300** includes a core **310**. The core **310** may be formed of one or more layers of glass fiber, resin, filler, pre-preg, epoxy, silica filler, Ajinomoto Build-up Film (ABF), polyimide, molding compound, other materials, and/or combinations thereof. In some embodiments, for example, two layers are of material make up the core **310**. The core **310** may be formed of organic and/or inorganic materials. In some embodiments, the core **310** includes one or more passive components (not shown) embedded inside. The core **310** may comprise other materials or components. Conductive vias **320** are formed extending through the core **310**. The conductive vias **320** comprise a conductive material **320A** such as copper, a copper alloy, or other conductors, and may include a barrier layer (not shown), liner (not shown), seed layer (not shown), and/or a fill material **320B**, in some embodiments. The conductive vias **320** provide vertical electrical connections from one side of the core **310** to the other side of the core **310**. For example, some of the conductive vias **320** are coupled between conductive features at one side of the core **310** and conductive features at an opposite side of the core **310**. Holes for the conductive vias **320** may be formed using a drilling process, photolithography, a laser process, or other methods, as examples, and the holes of the conductive vias **320** are then filled or plated with conductive material. In some embodiments, the conductive vias **320** are hollow conductive through vias having centers that are filled with an insulating material. Redistribution structures **340A** and **340B** are formed on opposing sides of the core **310**. The redistribution structures **340A** and **340B** are electrically coupled by the conductive vias **320**, and fan-in/fan-out electrical signals.

[0083] The redistribution structures **340A** and **340B** each include dielectric layers, formed of ABF,

pre-preg, or the like, and metallization patterns. Each respective metallization pattern has line portions on and extending along a major surface of a respective dielectric layer, and has via portions extending through the respective dielectric layer. The redistribution structures **340A** and **340B** each, respectively, include under-bump metallurgies (UBMs) **330A** and **330B** for external connection, and solder resists **350A** and **350B** protecting the features of the redistribution structures **340A** and **340B**. The redistribution structure **340A** is attached to the redistribution structure **200** by the UBMs **330A** through the conductive connectors **390** as illustrated in FIG. **24**. More or fewer dielectric layers and metallization patterns may be formed in the redistribution structures **340A** and **340B** than shown in FIG. **24**.

[0084] The core substrate **300** may include active and passive devices (not shown), or may be free from either active devices, passive devices, or both. A wide variety of devices such as transistors, capacitors, resistors, inductors, combinations of these, and the like may be used. The devices may be formed using any suitable methods.

[0085] Conductive connectors **390** may be used to bond the core substrates **300A** and **300B** to the redistribution structure **200** as illustrated in FIG. **18**. The conductive connectors **390** may be first formed on either of the core substrates **300A** and **300B**, or on the conductive pads **214** of the redistribution structure **200**, and then reflowed to complete the bond. For example, in the embodiment shown in FIG. **24**, conductive connectors **390** are formed on UBMs **330A** of the bottom redistribution structure **340A** with a pitch in a range of about 20 μm to about 500 μm . The conductive connectors **390** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **390** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. Integration of such conductive connectors **390** may provide flexibility in placement for semiconductor devices, such as integrated passive device (IPD) chips, integrated voltage regulators (IVRs), active chips, among other electrical components, to implement system-on-a-chip type of package components, thus reducing fabrication complexity. Such embodiments may also provide a greater amount of flexibility for various other package configurations as well.

[0086] In some embodiments, the conductive connectors **390** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the conductive connectors **390** comprise metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process.

[0087] In some embodiments, the core substrate **300** may be placed on the redistribution structure **200** using a pick and place process or another suitable process and the conductive connectors **390** bonded by flip chip bonding process or other suitable bonding process. In some embodiments, the conductive connectors **390** are reflowed to attach the core substrates **300A** and **300B** to the redistribution structure **200** by way of conductive pads **214**. The conductive connectors **390** electrically and/or physically couple the core substrate **300** to the redistribution structure **200** and through the local interconnect components **120** to integrated circuit dies **512**, **514**, and **516**. The double-sided connection of the integrated circuit dies **512**, **514**, and **516** to each other and to the core substrate **300** through the local interconnect component **120** may increase the communication bandwidth between the integrated circuit dies while maintaining low contact resistance and high reliability, which may achieve high signal integrity and power integrity. Power distribution

networks with two-sided connections through the local interconnect components may be managed without a significant voltage drop.

[0088] The conductive connectors **390** may have an epoxy flux (not shown) formed thereon before they are reflowed with at least some of the epoxy portion of the epoxy flux remaining after the core substrate **300** is attached to the redistribution structure **200**.

[0089] In some embodiments, an underfill **380**, as shown in FIG. **24**, is formed surrounding the conductive connectors **390** between the redistribution structure **200** and the core substrate **300**. The underfill **380** may reduce stress and protect the joints resulting from the reflowing of the conductive connectors **390**. The underfill **380** may be formed by a capillary flow process after the redistribution structure **200** is attached, or may be formed by a suitable deposition method. In some embodiments, a single layer of underfill **380** is formed beneath multiple adjacent devices, and further subsequent underfills (not shown) or encapsulants (not shown) may be formed beneath and/or around additional devices placed on top of the singulated package component **100**.

[0090] External connectors **620**, as shown in FIG. **24**, are formed on the UBMs **330B** of the core substrate **300**. The external connectors **620** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, or the like. The external connectors **620** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the external connectors **620** are formed by initially forming a layer of reflowable material on the UBMs **330** through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of reflowable material has been formed on the UBMs **330B** a reflow may be performed in order to shape the material into the desired bump shapes.

[0091] Embodiments may achieve advantages. For example, embedded double-sided local interconnect component may increase the communication bandwidth between the integrated circuit dies while maintaining low contact resistance and high reliability. The embedded double-sided local interconnect component may increase the communication bandwidth between the integrated circuit dies while maintaining low contact resistance and high reliability. The double-sided local interconnect connections may achieve high signal integrity and power integrity, and may enable power distribution networks to be managed through the two-sided connection without a significant voltage drop. Forming double-sided local interconnect components without adhesives such as die attach films may allow the reliability window to be enlarged. The formation of compound cracks between attached integrated circuit dies may be reduced by omitting a grinding process step to remove the adhesive, improving device reliability.

[0092] In accordance with an embodiment, a semiconductor structure includes: a first redistribution structure, the first redistribution structure including a first redistribution layer, the first redistribution layer including a first dielectric layer and a first metallization layer; a first local interconnect component disposed on the first redistribution structure, sidewalls of the first local interconnect component being surrounded by an encapsulant, the first local interconnect component including a first plurality of redistribution layers, the first plurality of redistribution layers including a first plurality of conductive features on a first side of the first local interconnect component, the first side physically contacting the first redistribution layer, each of the first plurality of conductive features being physically and electrically coupled to respective conductive features of the first metallization layer; and a first interconnect structure over a second side of the first local interconnect component, the second side being opposite the first side, the first interconnect structure including a second plurality of conductive features and a third plurality of conductive features, the second plurality of conductive features being electrically coupled to the third plurality of conductive features through the first local interconnect component. In an embodiment, the semiconductor structure further includes: a first integrated circuit die physically and electrically coupled to the first interconnect structure opposite the first local interconnect component, the first integrated circuit die being electrically coupled to the second plurality of

conductive features of the first interconnect structure; and a second integrated circuit die physically and electrically coupled to the first interconnect structure opposite the first local interconnect component, the second integrated circuit die being adjacent to the first integrated circuit die, the second integrated circuit die being electrically coupled to the third plurality of conductive features of the first interconnect structure. In an embodiment, the semiconductor structure further includes: a plurality of through vias extending through the encapsulant, a bottom surface of each through via of the plurality of through vias being coupled to a respective conductive feature of the first redistribution structure, a top surface of each through via of the plurality of through vias being coupled to a respective conductive feature of the first interconnect structure. In an embodiment, the semiconductor structure further includes a semiconductor package attached to the first redistribution structure opposite the first local interconnect component, the semiconductor package being coupled to the first redistribution structure by a plurality of conductive connectors. In an embodiment, the semiconductor package is a core substrate. In an embodiment, the semiconductor structure further includes: the semiconductor package is electrically coupled to the first interconnect structure through the first local interconnect component. In an embodiment, the semiconductor structure further includes: the first local interconnect component includes a first metal pillar, the first metal pillar physically contacting a bottom surface of the first interconnect structure, sidewalls of the first metal pillar being covered by the encapsulant. In an embodiment, the semiconductor structure further includes: the first plurality of conductive features of the first local interconnect component includes a second metal pillar, a bottom surface of the second metal pillar physically contacting the first redistribution structure. In an embodiment, the semiconductor structure further includes: the first local interconnect component further includes a dielectric material, sidewalls of the second metal pillar being covered by the dielectric material.

[0093] In accordance with another embodiment, a semiconductor device includes: a back-side redistribution structure, the back-side redistribution structure including a first metallization pattern and a first dielectric layer; a first local interconnect component, a bottom side of the first local interconnect component being attached to a top side of the back-side redistribution structure, a first plurality of conductive features being on the bottom side of the first local interconnect component, the first plurality of conductive features being physically and electrically coupled to the first metallization pattern; a first encapsulant on the back-side redistribution structure, the first encapsulant covering sidewalls of the first local interconnect component; a first integrated circuit die on the first encapsulant, the first integrated circuit die over a first portion of the first local interconnect component, the first integrated circuit die being electrically coupled to the first local interconnect component; and a second integrated circuit die on the first encapsulant, the second integrated circuit die over a second portion of the first local interconnect component, the second integrated circuit die being electrically coupled to the first integrated circuit die through the first local interconnect component. In an embodiment, the semiconductor device further includes a first through via and a second through via on the back-side redistribution structure, the first through via and the second through via extending through the first encapsulant, the first through via being electrically coupled to the first integrated circuit die, the second through via being coupled to the second integrated circuit die. In an embodiment, the semiconductor device further includes a first through via and a second through via on the back-side redistribution structure, the first through via and the second through via extending through the first encapsulant, the first through via being electrically coupled to the first integrated circuit die, the second through via being coupled to the second integrated circuit die. In an embodiment, the semiconductor device further includes an interconnect structure over the first local interconnect component and the first encapsulant, the first integrated circuit die and the second integrated circuit die attached to the interconnect structure opposite the first local interconnect component. In an embodiment, the semiconductor device further includes a third integrated circuit die attached to the interconnect structure neighboring the first integrated circuit die. In an embodiment, the semiconductor device further includes a second

local interconnect component, a bottom side of the second local interconnect component being attached to a top side of the back-side redistribution structure, the first encapsulant covering sidewalls of the second local interconnect component, a first portion of the second local interconnect component being under the first integrated circuit die, a second portion of the second local interconnect component being under the third integrated circuit die. In an embodiment, the semiconductor device further includes the third integrated circuit die being electrically coupled to the first integrated circuit die through the second local interconnect component. In an embodiment, the semiconductor device further includes a semiconductor package attached to the back-side redistribution structure opposite the interconnect structure, the second integrated circuit die being electrically coupled to the semiconductor package through the first local interconnect component, the third integrated circuit die being electrically coupled to the semiconductor package through the second local interconnect component.

[0094] In accordance with yet another embodiment, a method of forming a semiconductor device includes: forming a first plurality of redistribution layers on a first substrate; forming a first plurality of connectors on the first plurality of redistribution layers; singulating the first substrate, the first plurality of redistribution layers, and the first plurality of connectors into a plurality of local interconnect structures, a first local interconnect structure of the plurality of local interconnect structures including: a singulated portion of the first substrate; a singulated portion of the first plurality of redistribution layers on the singulated portion of the first substrate; and a second plurality of connectors on the singulated portion of the first plurality of redistribution layers, the second plurality of connectors being a subset of the first plurality of connectors; attaching the first local interconnect structure to a second substrate; forming a first encapsulant over the first local interconnect structure and the second substrate; removing a top portion of the first encapsulant and removing the singulated portion of the first substrate from the first local interconnect structure; forming a first redistribution structure over a first side of the first local interconnect structure and the first encapsulant; removing the first local interconnect structure, the first encapsulant, and the first redistribution structure from the second substrate and attaching the first redistribution structure to a third substrate; forming a second redistribution structure over a second side of the first local interconnect structure and the first encapsulant, the second side being opposite the first side; and attaching a first integrated circuit die and a second integrated circuit die to the second redistribution structure, the first integrated circuit die being electrically coupled to the second integrated circuit die through the first local interconnect structure. In an embodiment, forming the first plurality of connectors includes forming a plurality of metal pillars on the first plurality of redistribution layers. In an embodiment, the method further includes attaching a semiconductor package to the first redistribution structure opposite the second redistribution structure, the semiconductor package being electrically coupled to the first local interconnect structure through the first redistribution structure.

[0095] Embodiments disclosed herein may provide for a method including forming a first plurality of redistribution layers on a first substrate, forming a first plurality of connectors on the first plurality of redistribution layers, and singulating the first substrate, the first plurality of redistribution layers, and the first plurality of connectors into a plurality of local interconnect structures. A first local interconnect structure of the plurality of local interconnect structures comprises: a singulated portion of the first substrate, a singulated portion of the first plurality of redistribution layers on the singulated portion of the first substrate, and a second plurality of connectors on the singulated portion of the first plurality of redistribution layers, the second plurality of connectors being a subset of the first plurality of connectors. The method further includes attaching the first local interconnect structure to a second substrate, forming a first encapsulant over the first local interconnect structure and the second substrate, removing a top portion of the first encapsulant and removing the singulated portion of the first substrate from the first local interconnect structure, forming a first redistribution structure over a first side of the first

local interconnect structure and the first encapsulant, removing the first local interconnect structure, the first encapsulant, and the first redistribution structure from the second substrate and attaching the first redistribution structure to a third substrate. The method further includes forming a second redistribution structure over a second side of the first local interconnect structure and the first encapsulant, the second side being opposite the first side, and attaching a first integrated circuit die and a second integrated circuit die to the second redistribution structure, the first integrated circuit die being electrically coupled to the second integrated circuit die through the first local interconnect structure.

[0096] Other embodiments disclosed herein may provide for a method including forming on a first carrier substrate a plurality of local interconnect structures, each local interconnect structure including a stack of metallization lines embedded within respective dielectric layers of a stack of dielectric layers, singulating the plurality of local interconnect structures into individual local interconnect structures, forming conductive vias adjacent the local interconnect structure, encapsulating the individual local interconnect structure and the adjacent conductive vias in an encapsulant, forming a back-side redistribution structure electrically contacting the local interconnect structure and extending over a back-side of the encapsulant, forming a front-side redistribution structure electrically contacting the local interconnect structure and extending over a front-side of the encapsulant, and bonding an integrated circuit package to the front-side redistribution structure, wherein the integrated circuit package includes a first integrated circuit and a second integrated circuit further wherein the first integrated circuit is in electrical communication with the second integrated circuit through the individual local interconnect structure.

[0097] Yet other embodiments disclosed herein may provide for a method including forming a first local interconnect component and a conductive via surrounded by an encapsulant, the first local interconnect component comprising a first plurality of redistribution layers, the first plurality of redistribution layers comprising a first plurality of conductive features on a first side of the first local interconnect component, the first side physically contacting the first redistribution layer, forming a first redistribution structure on a first side of the encapsulant and being in electrical contact with external electrical connectors of the first local interconnect. The first redistribution structure comprises a first redistribution layer, the first redistribution layer comprising a first dielectric layer and a first metallization layer. The method further includes forming a second redistribution structure on a second side of the encapsulant, opposite the first side of the encapsulant, the second redistribution structure being in electrical communication with the local interconnect structure, the conductive via providing an electrical communication path between the first redistribution structure and the second redistribution structure, and bonding an integrated circuit package to the second redistribution structure, the integrated circuit package including a first integrated circuit and a second integrated circuit, wherein the local interconnect structure provides an electrical communication path between the first integrated circuit and the second integrated circuit.

[0098] Certain aspects of embodiments disclosed herein may provide for a device including a local interconnect structure a stack of metallization lines embedded within respective dielectric layers of a stack of dielectric layers. The device also includes a conductive via adjacent the local interconnect structure. The device also includes an encapsulant encapsulating the local interconnect structure and the conductive via. The device also includes a back-side redistribution structure electrically contacting the local interconnect structure and extending over a back-side of the encapsulant. The device also includes a front-side redistribution structure electrically contacting the local interconnect structure and extending over a front-side of the encapsulant. The device also includes an integrated circuit package electrically and physically contacting the front-side redistribution structure, where the integrated circuit package includes a first integrated circuit and a second integrated circuit, and further where the first integrated circuit is in electrical communication with the second integrated circuit through the stack of metallization layers of the

local interconnect structure.

[0099] Different aspects of embodiments disclosed herein may provide for a device including a redistribution structure including a front-side region, a back-side region, and a local interconnect region interjacent the front-side region and the back-side region. The device also includes an integrated circuit package including a first integrated circuit having first contact pads and a second integrated circuit having second contact pads, the integrated circuit package having a first solder joint with the front-side region of the redistribution structure a core substrate having a second solder joint with the back-side region of the redistribution structure. The device also includes the local interconnect region of the redistribution structure including a local interconnect structure electrically connecting the first integrated circuit to the second integrated circuit, an encapsulant laterally surrounding the local interconnect structure, and a conductive via extending through the encapsulant and electrically connecting the front-side region of the redistribution structure to the back-side region of the redistribution structure. The back-side region includes a topmost dielectric layer forming a dielectric-to-dielectric bond with a bottommost dielectric layer of the local interconnect region, and further including a topmost metal contact forming a metal-to-metal bond with a bottommost metal contact of the local interconnect region.

[0100] Other aspects of embodiments disclosed herein may provide for a device including a local interconnect region including: a local interconnect structure, the local interconnect structure including first contact pads on a topmost surface, second contact pads on a bottommost surface, and electrical interconnections between respective first contact pads and second contact pads, an encapsulant laterally surrounding the local interconnect structure, and a conductive via extending from a topmost surface of the encapsulant to a bottommost surface of the encapsulant. The device also includes a front-side redistribution region including bottommost metal pads forming respective metal-to-metal bonds with respective first contact pads of the local interconnect structure. The device also includes a back-side redistribution region including topmost metal pads forming respective metal-to-metal bonds with respective second contact pads of the local interconnect structure. The device also includes a package component physically and electrically contacting the front-side redistribution region, the package component including a first integrated circuit die and a second integrated circuit die, the first integrated circuit die being electrically connected to the second integrated circuit die through the front-side redistribution region and the local interconnect structure. The device also includes a substrate physically and electrically contacting the back-side redistribution region.

[0101] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device comprising: a local interconnect structure a stack of metallization lines embedded within respective dielectric layers of a stack of dielectric layers; a conductive via adjacent the local interconnect structure; an encapsulant encapsulating the local interconnect structure and the conductive via; a back-side redistribution structure electrically contacting the local interconnect structure and extending over a back-side of the encapsulant; a front-side redistribution structure electrically contacting the local interconnect structure and extending over a front-side of the encapsulant; and an integrated circuit package electrically and physically contacting the front-side

redistribution structure, wherein the integrated circuit package includes a first integrated circuit and a second integrated circuit, and further wherein the first integrated circuit is in electrical communication with the second integrated circuit through the stack of metallization layers of the local interconnect structure.

2. The device of claim 1, wherein the local interconnect structure includes a first metal bond pad, the front-side redistribution structure includes a second metal bond pad, and further wherein the first metal bond pad and the second metal bond pad form a metal-to-metal bond.

3. The device of claim 2, wherein the local interconnect structure includes a third metal bond pad on an opposite of the local interconnect structure relative to the first metal bond pad, the back-side redistribution structure includes a fourth metal bond pad, and further wherein the third metal bond pad and the fourth metal bond pad form a second metal-to-metal bond.

4. The device of claim 2, further comprising an underfill material surrounding the first metal bond pad, and a dielectric material, comprising a different material than the underfill material, surrounding the second metal bond pad.

5. The device of claim 2, further comprising a solder joint between the front-side redistribution structure and the integrated circuit package.

6. The device of claim 3, wherein a bottommost surface of the conductive via is co-planar with a bottommost surface of the third metal bond pad, and wherein a topmost surface of the conductive via is co-planar with a topmost surface of the first metal bond pad.

7. The device of claim 1, wherein the integrated circuit package includes a first encapsulant laterally surrounding the first integrated circuit and the second integrated circuit, wherein the device further includes a second encapsulant laterally surrounding the local interconnect structure and the conductive via, and wherein sidewalls of the first encapsulant and the second encapsulant are co-terminus.

8. The device of claim 1, wherein the local interconnect structure electrically connect the first integrated circuit to the second integrated circuit.

9. A device comprising: a redistribution structure including a front-side region, a back-side region, and a local interconnect region interjacent the front-side region and the back-side region; an integrated circuit package including a first integrated circuit having first contact pads and a second integrated circuit having second contact pads, the integrated circuit package having a first solder joint with the front-side region of the redistribution structure; a core substrate having a second solder joint with the back-side region of the redistribution structure; the local interconnect region of the redistribution structure including a local interconnect structure electrically connecting the first integrated circuit to the second integrated circuit, an encapsulant laterally surrounding the local interconnect structure, and a conductive via extending through the encapsulant and electrically connecting the front-side region of the redistribution structure to the back-side region of the redistribution structure; and the back-side region including a topmost dielectric layer forming a dielectric-to-dielectric bond with a bottommost dielectric layer of the local interconnect region, and further including a topmost metal contact forming a metal-to-metal bond with a bottommost metal contact of the local interconnect region.

10. The device of claim 9, wherein the core substrate includes a first redistribution layer solder bonded to the back-side region of the redistribution structure and comprising one or more conductive layers embedded within one or more respective dielectric layers, a second redistribution layer comprising one or more second conductive layers embedded within one or more respective second dielectric layers, a core interjacent the first redistribution layer and the second redistribution layer, and one or more conductive vias extending through the core and electrically connecting the first redistribution layer and the second redistribution layer.

11. The device of claim 9, wherein the bottommost surface of the front-side region of the redistribution structure extends is above the topmost surface of the local interconnect region of the redistribution structure.

- 12.** The device of claim 9, wherein the integrated circuit package includes a second encapsulant, the second encapsulant being separated from the encapsulant of the local interconnect region of the redistribution structure by the front-side region of the redistribution structure.
- 13.** The device of claim 12, wherein sidewalls of the encapsulant and sidewalls of the second encapsulant are coterminous.
- 14.** The device of claim 9, wherein the local interconnect structure overlaps with the first integrated circuit and the second integrated circuit when viewed in a top-down perspective.
- 15.** The device of claim 9, further comprising an underfill material, different than the encapsulant, surrounding the first solder joint.
- 16.** The device of claim 9, wherein the integrated circuit package includes a third integrated circuit, and further comprising a second local interconnect structure electrically connecting the first integrated circuit and the third integrated circuit.
- 17.** A device comprising: a local interconnect region including: a local interconnect structure, the local interconnect structure including first contact pads on a topmost surface, second contact pads on a bottommost surface, and electrical interconnections between respective first contact pads and second contact pads, an encapsulant laterally surrounding the local interconnect structure, and a conductive via extending from a topmost surface of the encapsulant to a bottommost surface of the encapsulant; a front-side redistribution region including bottommost metal pads forming respective metal-to-metal bonds with respective first contact pads of the local interconnect structure; a back-side redistribution region including topmost metal pads forming respective metal-to-metal bonds with respective second contact pads of the local interconnect structure; a package component physically and electrically contacting the front-side redistribution region, the package component including a first integrated circuit die and a second integrated circuit die, the first integrated circuit die being electrically connected to the second integrated circuit die through the front-side redistribution region and the local interconnect structure; and a substrate physically and electrically contacting the back-side redistribution region.
- 18.** The device of claim 17, further comprising a bottommost dielectric layer of the local interconnect structure forming a dielectric-to-dielectric bond with a topmost layer of the back-side redistribution region.
- 19.** The device of claim 17, wherein the local interconnect structure overlaps with the first integrated circuit die and the second integrated circuit die when viewed from a top-down perspective.
- 20.** The device of claim 17, wherein the second integrated circuit die comprises a vertically aligned stack of integrated circuits die.
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